

# G294-S45-AAP2

HPC/AI Server - Intel® Xeon® 6700/6500 - 2U DP 8 x PCIe Gen5 GPUs



## Key Features

- Supports up to 8 x Dual slot Gen5 GPUs
- Supports 4 x NVIDIA H200 NVL PCIe GPUs
- Dual Intel® Xeon® 6700/6500-Series Processors
- 8-Channel DDR5 RDIMM / MRDIMM, 16 x DIMMs
- 2 x 10 Gb/s LAN ports via Intel® X710-AT2
- 8 x 2.5" SATA/SAS-4 hot-swap bays
- 8 x FHFL PCIe Gen5 x16 for GPUs
- 2 x LP PCIe Gen5 x16
- Dual 3000 W 80 PLUS Titanium redundant power supply

## Applications

AI Platform, AI Training Server, AI Inference Server, Visual Computing, High-Performance Computing Server...

## Specification

|                         |                                                                                                                                                                                                                                                                                      |                              |                                                                                                                                                                  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dimensions</b>       | 2U (W448 x H87.5 x D962 mm, depth incl. 112 mm rear fans)                                                                                                                                                                                                                            | <b>RAID</b>                  | Require RAID add-in cards<br>Onboard VROC key headers                                                                                                            |
| <b>Motherboard</b>      | MS44-G21                                                                                                                                                                                                                                                                             | <b>PCIe Expansion Slots</b>  | 8 x FHFL PCIe Gen5 x16, for GPUs<br>2 x LP PCIe Gen5 x16<br><br>*The system supports 4 x NVIDIA H200 NVL PCIe GPUs at 25°C ambient, arranged as four 1-GPU sets. |
| <b>CPU</b>              | Intel® Xeon® 6 Processors<br>- Intel® Xeon® 6700-Series Processors<br>- Intel® Xeon® 6500-Series Processors<br>Dual processor, TDP up to 250 W<br>- At 30°C ambient, TDP up to 300 W<br>- At 25°C ambient, TDP up to 350 W                                                           | <b>I/O Ports</b>             | <b>Rear:</b><br>2 x USB 3.2 Gen1, 1 x VGA, 2 x RJ45, 1 x MLAN                                                                                                    |
| <b>OS Compatibility</b> | <a href="#">OS Support List</a>                                                                                                                                                                                                                                                      | <b>TPM</b>                   | 1 x TPM header (SPI), Optional TPM 2.0 kit: CTM012<br>1 x PRoT connector (only enabled on RoT SKU)                                                               |
| <b>Socket</b>           | 2 x LGA 4710 (Socket E2)                                                                                                                                                                                                                                                             | <b>Power Supply</b>          | Dual 3000 W 80 PLUS Titanium redundant power supply<br>AC Input: 100-240 V<br><br>*The system power supply requires C19 power cord.                              |
| <b>Chipset</b>          | System on Chip                                                                                                                                                                                                                                                                       | <b>System Fans</b>           | 8 x 80x80x80mm                                                                                                                                                   |
| <b>Memory</b>           | 8-Channel DDR5 RDIMM/MRDIMM, 16 x DIMMs<br>[RDIMM] Up to 6400 MT/s<br>[MRDIMM] Up to 8000 MT/s <sup>[1]</sup><br><br><sup>[1]</sup> MRDIMMs are supported only on select Intel® Xeon® 6 processors with P-cores.                                                                     | <b>Operating Properties</b>  | Operating: 10°C to 35°C, 8% to 80% (non-condensing)<br>Non-operating: -40°C to 60°C, 20% to 95% (non-condensing)                                                 |
| <b>LAN</b>              | 2 x 10 Gb/s LAN (1 x Intel® X710-AT2), Support NCSI<br>1 x 10/100/1000 Mb/s Management LAN                                                                                                                                                                                           | <b>Packaging Content</b>     | 1 x G294-S45-AAP2<br>2 x CPU heatsinks<br>4 x GPU power cables (16-pin, 250mm)<br>4 x GPU power cables (16-pin, 280mm)<br>4 x Carriers<br>1 x L-shape Rail kit   |
| <b>Video</b>            | Integrated in ASPEED® AST2600<br>- 1 x VGA port                                                                                                                                                                                                                                      | <b>Ordering Part Numbers</b> | 6NG294S45DR000AAP2*                                                                                                                                              |
| <b>Storage</b>          | <b>Front hot-swap:</b><br>4 x 2.5" <b>Gen5 NVMe/SATA/SAS-4</b> <sup>[1]</sup><br>4 x 2.5" <b>SATA/SAS-4</b> <sup>[1]</sup><br><br><sup>[1]</sup> SAS card is required to support SAS drives.<br><sup>[2]</sup> NVMe support requires additional NVMe cables and occupies PCIe slots. | <b>Optional Parts</b>        | - NVMe cables (A)x2+(B):<br>(A) MCIO 4i to MCIO 4i: 25CF9-8007A4-A4R<br>(B) MCIO 8i to 2 x MCIO 4i: 25CF9-450703-A4R                                             |



Learn more at: <https://www.gigabyte.com/Enterprise>

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